

Sharp Swap-Polynomial Degrees on M_2 and an Exterior-Volume Construction in Every Dimension

Automated mathematics run `fa_banach_001`, lane 7

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Status. Substantial partial result, likely valid, pending human review. The lowest swap degree is determined exactly for 2×2 matrices: it is 4 for ordinary tensor polynomials and 3 when traces are allowed. In general dimension, a new direct trace-swap construction has degree $d^2 - 1$. Sharp minimality for $d \geq 3$ remains open.

1 Source question and answer

Let $\Gamma \in M_d(\mathbb{C})^{\otimes 2}$ be the swap tensor,

$$\Gamma = \sum_{i,j=1}^d E_{ij} \otimes E_{ji}.$$

A (trace) tensor polynomial P is a *swap polynomial* if $P(X) = c(X)\Gamma$ for a scalar polynomial c which is not identically zero. Degree below always means total matrix-variable degree; traces carry their usual word degree. Huber asks for the lowest possible degree, both for ordinary tensor polynomials and when trace polynomials are allowed [1, p. 19]:

Second, we recall that Amitsur and Levitzki showed that M_d satisfies the standard identity s_{2d} in $2d$ variables while it does not satisfy any polynomial identity of lower degree. The corresponding question on the lowest degree for central polynomials, swap polynomials, and more general polynomials on tensor product spaces is unresolved [36]. Similarly, we do not know what is the lowest degree required when trace polynomials instead of only polynomials are allowed. To approach this problem, a more direct construction for permutation polynomials and tensor polynomial identities than the ones presented here is desirable.

Figure 1: The degree question in the source paper [1, p. 19].

Write μ_d for the least ordinary swap degree and μ_d^{tr} for the least trace-swap degree.

Theorem 1 (Sharp 2×2 answer and a general construction). *Over $\mathbb{C}$ one has*

$$\boxed{\mu_2 = 4, \quad \mu_2^{\text{tr}} = 3.}$$

More generally, for every $d \geq 2$ there is a nonzero multilinear trace-swap polynomial of degree $d^2 - 1$. Consequently

$$3 \leq \mu_d^{\text{tr}} \leq d^2 - 1.$$

An exact Schur–Weyl rank computation additionally gives $5 \leq \mu_3^{\text{tr}} \leq 8$.

Thus allowing traces strictly lowers the minimum in dimension two. The all-dimensional upper bound is obtained by one coordinate-free formula, rather than by clearing a Gram determinant.

2 The exterior-volume trace-swap formula

Put $N = d^2 - 1$ and $W = \mathfrak{sl}_d$. The trace pairing $\langle X, Y \rangle = \text{tr}(XY)$ is nondegenerate on W . Choose a nonzero volume form $\Omega \in \bigwedge^N W^*$ and, for $X_1, \dots, X_N \in M_d(\mathbb{C})$, put

$$x_i = X_i - \frac{\text{tr } X_i}{d} I, \quad \Delta(X_1, \dots, X_N) = \Omega(x_1, \dots, x_N).$$

For each i , define the cofactor matrix $x_i^\#$ by

$$\text{tr}(Zx_i^\#) = \Omega(x_1, \dots, x_{i-1}, Z, x_{i+1}, \dots, x_N) \quad (Z \in W). \quad (1)$$

Nondegeneracy makes $x_i^\#$ unique. It is homogeneous of degree $N - 1$.

Proposition 1 (Exterior-volume identity). *For all $X_1, \dots, X_N \in M_d(\mathbb{C})$,*

$$\boxed{\frac{\Delta}{d} I \otimes I + \sum_{i=1}^N x_i \otimes x_i^\# = \Delta \Gamma.} \quad (2)$$

Both sides are multilinear of degree $N = d^2 - 1$, and the left side is a tensor trace polynomial.

Proof. If $\Delta \neq 0$, the matrices $x_1, \dots, x_N$ form a basis of W . Equation (1) gives

$$\text{tr}(x_j x_i^\#) = \Delta \delta_{ij}.$$

Hence the trace-dual basis to $\{I, x_1, \dots, x_N\}$ is $\{I/d, x_1^\#/\Delta, \dots, x_N^\#/\Delta\}$. For every basis $\{e_a\}$ of M_d and its trace-dual basis $\{e^a\}$, the canonical dual-basis tensor satisfies

$$\sum_a e_a \otimes e^a = \sum_{i,j} E_{ij} \otimes E_{ji} = \Gamma.$$

Multiplication by Δ proves (2) when $\Delta \neq 0$, and polynomial continuation proves it everywhere.

It remains to justify the word “trace.” Conjugation acts on W through the adjoint representation, whose determinant is one. Thus Ω , and hence Δ , is invariant under simultaneous conjugation. The first fundamental theorem for matrix invariants expresses Δ as a polynomial in traces of words [2]; the source recalls precisely this trace-polynomial description of equivariants and invariants [1, Sections II and VI]. Taking the cyclic derivative of such a trace expression in its i th variable produces the matrix trace polynomial characterized by (1). Therefore every $x_i^\#$, and the whole left side of (2), is a tensor trace polynomial. Choosing the x_i to be any basis of W shows that Δ is not identically zero. $\square$

3 The cubic formula on M_2

For a 2×2 matrix A , write $A_0 = A - \frac{1}{2} \text{tr}(A)I$, and similarly define B_0, C_0 . Set

$$\tau = \text{tr}(A[B, C]).$$

Proposition 1, with the invariant volume form $\Omega(a, b, c) = \text{tr}(a[b, c])$ on $\mathfrak{sl}_2$, becomes the explicit four-term identity

$$\boxed{\frac{\tau}{2} I \otimes I + A_0 \otimes [B, C] + B_0 \otimes [C, A] + C_0 \otimes [A, B] = \tau \Gamma.} \quad (3)$$

Indeed $[B, C], [C, A], [A, B]$ are exactly the three trace-dual cofactors: for example $\text{tr}(A_0[B, C]) = \tau$, while $\text{tr}(B_0[B, C]) = \text{tr}(C_0[B, C]) = 0$. The coefficient is nonzero, for example on the three Pauli matrices. Thus $\mu_2^{\text{tr}} \leq 3$.

The following elementary observation proves sharpness and, in fact, gives a dimension-independent lower bound.

Lemma 1. *For every $d \geq 2$, no nonzero tensor trace swap polynomial has degree at most two.*

Proof. Taking homogeneous components and polarizing reduces the claim to multilinear degrees 0, 1, 2. A scalar coefficient of a multilinear swap polynomial is a simultaneous-conjugation invariant. In degree one it is a multiple of $\text{tr} A$. Evaluating at $A = I$, every tensor trace monomial is a multiple of $I \otimes I$, whereas a nonzero multiple of Γ is not; hence the coefficient vanishes. Degree zero is identical.

In degree two the scalar coefficient has the form

$$c(A, B) = u \text{tr}(A) \text{tr}(B) + v \text{tr}(AB).$$

Take $A = E_{11}$ and $B = E_{22}$. Every degree-two tensor trace monomial is diagonal in the product matrix-unit basis, while $u\Gamma$ has the nonzero off-diagonal summand $uE_{12} \otimes E_{21}$. Thus $u = 0$. Taking $A = B = I$ then makes the left side a multiple of $I \otimes I$ and the right side $vd\Gamma$, so $v = 0$. $\square$

Together with (3), the lemma proves $\mu_2^{\text{tr}} = 3$.

4 The ordinary degree on M_2

Procesi gives the following ordinary tensor polynomial [3, Section 3]:

$$\begin{aligned} P(x, y) &= I \otimes ([x, y]^2 + x[x, y]y) - y \otimes x[x, y] - x \otimes [x, y]y + xy \otimes [x, y] \\ &= [x, y]^2 \Gamma. \end{aligned} \tag{4}$$

For 2×2 matrices, the square of the traceless matrix $[x, y]$ is scalar, and it is not identically zero. Hence $\mu_2 \leq 4$.

For the reverse inequality, let $Q = \sum_i p_i \otimes q_i = c\Gamma$ be an ordinary swap polynomial. Applying the multiplication map $m(U \otimes V) = UV$ gives

$$\sum_i p_i q_i = dcI,$$

an ordinary central polynomial of the same degree. There is no nonzero central polynomial on M_2 of degree at most three. For completeness, after homogenization and polarization, only the trilinear case needs checking: write $q(A, B, C) = \sum_{\sigma \in S_3} a_\sigma A_{\sigma(1)} A_{\sigma(2)} A_{\sigma(3)}$. Requiring the two off-diagonal entries and the diagonal difference to vanish on the six tests

$$\begin{array}{l|l} (E_{11}, E_{11}, E_{11}) & 11 - 22 \\ (E_{11}, E_{11}, E_{12}) & 12 \\ (E_{11}, E_{11}, E_{21}) & 21 \\ (E_{11}, E_{12}, E_{11}) & 12 \\ (E_{11}, E_{12}, E_{21}) & 11 - 22 \\ (E_{11}, E_{21}, E_{11}) & 21 \end{array}$$

gives, in the order $ABC, ACB, BAC, BCA, CAB, CBA$, the coefficient matrix

$$\begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 \end{pmatrix}, \quad \det = -1.$$

Thus all a_σ vanish. In degree two, writing $q(A, B) = aAB + bBA$ and testing $(A, B) = (E_{11}, E_{12})$ and (E_{12}, E_{11}) gives $a = b = 0$; degree one is immediate. Consequently $\mu_2 \geq 4$, and (4) proves $\mu_2 = 4$.

5 Exact $d = 3$ check, limitations, and novelty boundary

For a multilinear trace degree r , let B be the permutations in S_{r+2} whose two output labels lie in distinct cycles; these encode tensor trace monomials. Let G consist of an input permutation followed by the output transposition; these encode scalar trace invariants times Γ . In the Schur–Weyl representation on $(\mathbb{C}^d)^{\otimes(r+2)}$, a trace swap exists exactly when the represented spans of B and G intersect nontrivially. The exact Gram entry is

$$\langle T(\pi), T(\sigma) \rangle_{\text{HS}} = d^{\#\text{cycles}(\pi^{-1}\sigma)}.$$

For $d = 3$, exact integer ranks are

r	rank B	rank G	rank $[B \ G]$	dim($\text{span } B \cap \text{span } G$)
3	58	6	64	0
4	319	23	342	0.

Together with Lemma 1, this proves $\mu_3^{\text{tr}} \geq 5$; Proposition 1 gives $\mu_3^{\text{tr}} \leq 8$. The accompanying script constructs these Gram matrices over the integers and also verifies (3) on all 4^3 matrix-unit triples. The $r = 4$ exact check is enabled by `--d3-exact`.

The full source question remains open here for $d \geq 3$. A proof that $d^2 - 1$ is always minimal, or a lower-degree construction, would require controlling the intersection above for a rapidly growing separated-output subspace. Eight focused attempts are documented in the attempt log.

A bounded novelty audit covered the run indexes, exact source phrases, arXiv searches for “swap polynomial” and “swap trace polynomial,” and the post-source paper of Procesi. The latter supplies (4) and higher-dimensional ordinary constructions, but no cubic trace identity or sharp trace minimum was found. Novelty confidence is moderate: the dual-basis proof is elementary once the invariant volume form is selected, so an uncatalogued prior occurrence remains possible.

Human review should focus on the trace-polynomial realization of the cofactors in Proposition 1, the degree convention relative to the source, and the exact-rank certificate for the optional $d = 3$ bound.

References

- [1] F. Huber, *Positive maps and trace polynomials from the symmetric group*, arXiv:2002.12887 (2020, revised 2021).
- [2] C. Procesi, *The invariant theory of $n \times n$ matrices*, Adv. Math. 19 (1976), 306–381.
- [3] C. Procesi, *A construction of swap or switch polynomials*, Adv. Math. 400 (2022), Article 108308; arXiv:2102.10657.